Laparoscopic Nissen Fundoplication: a minimal access alternative.

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The preferred treatment of gastroesophageal reflux has traditionally been Nissen fundoplication. This involves an extensive abdominal or thoracic incision and subsequently results in patient discomfort, an extended recovery period, and increased overall costs. In the advent of laparoscopic advances surgical correction of symptoms of gastroesophageal reflux are now being offered through minimal access surgery. Increased patient satisfaction, decreased costs, and a quicker return to activities of daily living, suggest why laparoscopic Nissen fundoplication, (LNF) is fast becoming the preferred alternative to correction of gastroesophageal reflux disease. This article will review gastroesophageal reflux and describe one surgical method of laparoscopic correction. The role of the perioperative nurse and implementation of the nursing process regarding this surgical procedure will be highlighted.

Comparative analysis of laparoscopic Nissen fundoplication and Rossetti modification in gastroesophageal reflux disease: A focus on life-quality enhancement.

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This study aims to investigate the focus of surgical treatment of gastroesophageal reflux disease (GERD) on enhancing life quality beyond symptom relief. The comparison involves laparoscopic Nissen fundoplication and Rossetti modification techniques. Patients intolerant to or experiencing relapse after medical therapy underwent either standard Nissen procedure (Group 1, $n = 61$) or Rossetti modification (Group 2, $n = 42$). A disease-specific quality of life questionnaire for GERD was utilized for evaluating life quality preoperatively and 2 years postoperatively. Symptom scores and patient satisfaction were also assessed. Preoperatively, groups were similar in symptom duration, hiatal hernia presence, and DeMeester scores ($p = 0.127$, $p = 0.427$, and 0.584, respectively). Both groups exhibited a statistically significant increase in life quality postoperatively ($p < 0.001$), with no significant intergroup difference. Symptoms decreased after both surgeries, except for dysphagia and bloating. Bloating significantly increased in both groups after surgery ($p = 0.018$ and $p = 0.017$, respectively), and dysphagia increased significantly only in Group 2 ($p = 0.007$). The surgery refusal rate was significantly higher in Group 2 for similar preoperative symptoms ($p = 0.040$). Despite increased life quality scores, the combination of increased dysphagia and bloating in patients undergoing Rossetti modification resulted in a decreased satisfaction rate.

Comparing Recovery from Desflurane and Sevoflurane in Patients with Different Body Fat Percentages: A Randomized Controlled Trial.

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Increased body fat may influence the partition coefficients of inhaled anesthetics. We compared patient responses to desflurane and sevoflurane anesthesia, as measured by a quicker recovery and fewer complications, in patients with higher body fat percentages, not only obese people. This study included 120 patients. Participants were stratified into low or high body fat percentages groups using bioelectrical impedance analysis and were randomized 1:1 to receive desflurane or sevoflurane as an inhaled anesthetic, recorded as Low-Desflurane, Low-Sevoflurane, High- Desflurane, and High-Sevoflurane.

Recovery time, Riker sedation-agitation scale scores, and complications were recorded over 1 hour in the post-anesthesia care unit. A total of 106 patients were analyzed. There were no significant differences in the overall recovery time between the patient subgroups with higher and lower body fat percentages; in addition, there were no significant differences in the incidence of nausea, vomiting,dizziness, or headache (all p>0.05). However, the incidence of agitation emergence in the HighSevoflurane subgroup was significantly higher compared to the High-Desflurane subgroup (33.3% vs.7.41%; p = 0.043). In conclusion, for patients with a lower body fat percentage, both desflurane and sevoflurane can provide good and fast recovery; for patients with a higher body fat percentage,desflurane may provide better recovery with a lower incidence of agitation emergence compared to sevoflurane.

Desflurane versus sevoflurane anesthesia and postoperative recovery in older adults undergoing minor- to moderate-risk noncardiac surgery - A prospective, randomized, observer-blinded, clinical trial.

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The effect of volatile anesthetics on postoperative recovery in older adults is still not entirely clear. Thus, we evaluated the effect of desflurane versus sevoflurane anesthesia on speed of postoperative recovery in older adults eligible for same-day discharge. We further evaluated the incidence of postoperative nausea and vomiting (PONV), bispectral index (BIS) values, and S100B concentrations. Single-center, prospective, observer-blinded, randomized clinical trial. Operating room. 190 patients ≥65 years of age and scheduled for minor- to moderate-risk noncardiac surgeries. Goal-directed administration of desflurane versus sevoflurane for maintenance of anesthesia with an intraoperative goal of BIS 50 ± 5. The primary outcome was the time to anesthesia recovery, which was defined as the time between arrival at the post-anesthesia care unit (PACU) and reaching criteria for discharge from PACU, based on modified Aldrete score ≥ 12 points. Modified Aldrete scores were assessed at PACU arrival and thereafter in five-minute intervals. PONV was evaluated during PACU stay and the first three postoperative days, BIS values were recorded during PACU stay, and S100B values were measured before and after surgery, and on the second postoperative day. 95 patients were randomized to receive desflurane, and 95 patients to receive sevoflurane. We did not observe a significant difference in median duration of postoperative recovery between the groups (desflurane: 0 min [0;0]; sevoflurane: 0 min [0;0]; p = 0.245). 77 patients (81.1%) in the desflurane group and 84 patients (88.4%) in the sevoflurane group already had Aldrete scores ≥12 points upon arrival at PACU (p = 0.277). There was also no significant difference in the incidences of PONV (p = 0.606), postoperative BIS values (p = 0.197), and postoperative maximum S100B concentrations (p = 0.821) between the groups. Despite previous reports, we did not observe significant faster recovery times after desflurane anesthesia. Both volatile anesthetics may be appropriate for same-day discharge in older adults.

Analysis Comparing the Recovery of Airway Reflexes and Cognitive Ability After Sevoflurane with Desflurane Anesthesia.

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Although sevoflurane and desflurane have nearly identical blood-gas solubilities, current research suggests that airway reflexes recover more quickly with desflurane than sevoflurane; however, cognitive function recovery varies substantially. The current study was piloted to appraise the lengths of time needed to recover from anesthesia following desflurane and sevoflurane anesthesia. A prospective clinical trial was piloted among 70 adult non-obese subjects who underwent elective surgery and were classified I-II by the "American Association of Anesthesiologists (ASA)". Sevoflurane and desflurane were tested among the subjects who were equally distributed. These agents were used in accordance with a normal general anaesthesia procedure. After they were extubated, tests for regaining cognitive function and airway reflexes were carried out, and different time intervals were recorded. The observations were calculated and $P < 0.05$ was used to conduct the statistical analysis. The average amount of time that passed between the patient's first vocal response and their first successful completion of the swallowing test was analogous between the two groups (T2) with 5.25 ± 3.11 vs 5.01 ± 2.12 in sevoflurane and desflurane, respectively. There was no significant variance at T2. For all the other time intervals of T1, T3, and T4, there was evidence of the significant variance.($P = 0.003$; 0.0013; <0.001, respectively). Desflurane causes patients to recover more quickly than sevoflurane does after laparoscopic cholecystectomy under controlled circumstances.

Differential effects of sevoflurane and desflurane on frontal intraoperative electroencephalogram dynamics associated with postoperative delirium.

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Intraoperative electroencephalogram (EEG) patterns associated with postoperative delirium (POD) development have been studied, but the differences in EEG recordings between sevoflurane- and desflurane-induced anesthesia have not been clarified. We aimed to distinguish the EEG characteristics of sevoflurane and desflurane in relation to POD development. We collected frontal four-channel EEG data during the maintenance of anesthesia from 148 elderly patients who received sevoflurane ($n = 77$) or desflurane ($n = 71$); 30 patients were diagnosed with delirium postoperatively. The patients were divided into four subgroups based on anesthetics and delirium status: sevoflurane delirium ($n = 17$), sevoflurane non-delirium ($n = 60$), desflurane delirium ($n = 13$), and desflurane non-delirium ($n = 58$). We compared spectral power, coherence, and pairwise phase consistency (PPC) between sevoflurane and desflurane, and between non-delirium and delirium groups for each anesthetic. In patients without POD, the sevoflurane non-delirium group exhibited higher EEG spectral power across 8.5-35 Hz (99.5% CI bootstrap analysis) and higher PPC from alpha to gamma bands ($p < 0.005$) compared to the desflurane non-delirium group. Conversely, in patients with POD, no significant EEG differences were observed between the sevoflurane and desflurane delirium groups. For the sevoflurane-induced patients, the sevoflurane delirium group had significantly lower power within 7.5-31.5 Hz (99.5% CI bootstrap analysis), reduced coherence over 8.9-23.8 Hz (99.5% CI bootstrap analysis), and lower PPC values in the alpha band ($p < 0.005$) compared with the sevoflurane non-delirium group. For the desflurane-induced patients, there were no significant differences in the EEG patterns between delirium and non-delirium groups. In normal patients without POD, sevoflurane demonstrates a higher power spectrum and prefrontal connectivity than desflurane. Furthermore, reduced frontal alpha power, coherence, and connectivity of intraoperative EEG could be associated with an increased risk of POD. These intraoperative EEG characteristics associated with POD are more noticeable in sevoflurane-induced anesthesia than in desflurane-induced anesthesia.

Comparison of maintenance and emergence characteristics after desflurane or

sevoflurane in outpatient anaesthesia.

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Both sevoflurane and desflurane have shorter emergence times compared to isoflurane based anaesthesia. Because of its pharmacological properties, desflurane appears to yield a rapid early and intermediate recovery compared with sevoflurane. The aim of this study was to assess the maintenance and emergence characteristics after anaesthesia with sevoflurane or desflurane. One hundred female patients scheduled to undergo daycare laparoscopic gynaecological surgery were enrolled for this prospective study. Patients were randomised into two groups to receive either desflurane [group I (D); n = 50] or sevoflurane [group II (S); n = 50] for maintenance of anaesthesia. The demographic data and the duration of procedure were comparable in both the groups. The early recovery time was shorter after maintenance of anaesthesia with desflurane compared with sevoflurane. However, this faster early recovery failed to lead to early readiness for home discharge. The intraoperative haemodynamic characteristics were comparable with both sevoflurane and desflurane. Both sevoflurane and desflurane provide a similar time to home readiness despite a faster early recovery with desflurane. The intraoperative haemodynamics are similar with both the agents.

[Newer inhalation anaesthetics and neuro-anaesthesia: what is the place for sevoflurane or desflurane?].

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The effects on cerebral circulation and metabolism of sevoflurane and desflurane are largely comparable to isoflurane. Both induce a direct vasodilation of the cerebral vessels, resulting in a less pronounced decrease in cerebral blood flow compared to the decrease in cerebral metabolism. This direct vasodilation seems to be dose-dependent and more pronounced for desflurane > isoflurane > sevoflurane. Many reports suggest luxury perfusion at high concentrations of desflurane. Sevoflurane maintains intact cerebral autoregulation up to 1.5 MAC. Desflurane induces a significant impairment in autoregulation, with a completely abolished autoregulation at 1.5 MAC. Both sevoflurane and desflurane (up to 1.5 MAC) maintain normal CO(2) regulation. As to their effect on final intracranial pressure (ICP), both sevoflurane and desflurane revealed no increases in ICP. However, compared to intravenous hypnotics, subdural ICP is higher with volatiles because of their tendency to increase cerebral swelling after dura opening (isoflurane > sevoflurane). Several case reports have noted seizure-like movements, as well as EEG recorded seizures during induction of sevoflurane anesthesia. Especially, in children during inhalational induction with hyperventilation at a high sevoflurane concentration, severe epileptiform EEG with a hyperdynamic response were observed, which urges for caution using inhalational sevoflurane induction in children for neurosurgical procedures. Neuroprotective properties (reduced neuronal death either by necrosis or apoptosis) have been attributed to all volatile agents. However, these neuroprotective effects have been described in experimental or animal models, so their possible effect on humans remains to be proven.

Sevoflurane. A review of its pharmacodynamic and pharmacokinetic properties and its clinical use in general anaesthesia.

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Sevoflurane is an ether inhalation general anaesthetic agent with lower solubility in blood than isoflurane or halothane but not desflurane. The low solubility and the absence of pungency facilitate rapid mask induction; the low blood solubility also expedites "wash-out' and therefore recovery from anaesthesia. Sevoflurane produces dose-dependent CNS, cardiovascular and respiratory depressant effects that generally parallel those of isoflurane. Sevoflurane is degraded by carbon dioxide absorbents to nephrontoxic (in rats) haloalkenes, although renal toxicity has not been observed in humans. Compared with other inhalation anaesthetics, negligible quantities of carbon monoxide are generated from degradation of sevoflurane by carbon dioxide absorbents. Sevoflurane has negligible airway irritant effects, which facilitates a "smooth' induction, even in comparison with halothane in paediatric patients, and makes sevoflurane especially amenable to rapid induction of anaesthesia in adults and children. Emergence, orientation an postoperative cognitive and psychomotor function recovery of paediatric outpatients is singnificantly more rapid from sevoflurane than from halothane anaesthesia. In adult inpatients and outpatients, emergence and orientation are significantly faster after sevoflurane than after isoflurane but not desflurane anaesthesia. Other recovery parameters (e.g. times to sitting, ambulation) occur at similar times after either sevoflurane or desflurane anaesthesia. Recovery of psychomotor function occurs at generally similar times after sevoflurane, isoflurane or desflurane. Compared with propofol, sevoflurane facilitates more predictable extubation times and significantly better postoperative modified Aldrete scores in outpatients, although cognitive and psychomotor recovery occurs at similar times for both agents. As a supplement to opioid anaesthesia during coronary bypass graft surgery or in those at risk for myocardial ischaemia, sevoflurane is comparable to isoflurane. Limited data suggest that it is also as useful as isoflurane for the maintenance of anaesthesia during neurosurgical or obstetric procedures. Sevoflurane is well tolerated by adult and paediatric patients during induction of anaesthesia, with a low incidence of mild airway complications (breath-holding, coughing, excitement and laryngospasm). During rapid induction, it is particularly better tolerated than isoflurane or halothane. Sevoflurane has a lower potential for hepatic injury than halothane. Unlike methoxyflurane, sevoflurane undergoes minimal intrarenal defluorination, which may account for the lack of fluoride ion-induced nephrotoxicity in humans, despite elevated plasma fluoride levels after its use. In summary, sevoflurane provides for a rapid and smooth induction of, and recovery from, anaesthesia. These features combined with its favourable cardiovascular profile should make sevoflurane the agent of choice for inhalation induction in adult and paediatric anaesthesia. Although further clinical evaluation will define the role of this agent relative to that of propofol and desflurane, sevoflurane should also prove to be a valuable alternative anaesthetic agent for adults in both outpatient and inpatient surgery.

[Critical evaluation of the new inhalational anesthetics desflurane and sevoflurane].

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New anaesthetic agents are being continuously developed to find the ideal agent. The most commonly used inhaled anaesthetic in adults is isoflurane and in children halothane. The need for, and the value of the new agents desflurane and sevoflurane depend on a comparison of the properties of a theoretically ideal agent with those of isoflurane, halothane and the new agents. Therefore, the following topics are discussed in the overview: pharmacokinetic properties, recovery parameters, mask induction in adults, paediatric anaesthesia, metabolism, stability in carbon dioxide absorbents and cardiovascular effects. Desflurane and sevoflurane may be considered a step toward the ideal inhalational agents. Both possess the advantage of rapid recovery from surgical anesthesia. Desflurane has a major advantage over sevoflurane: it is not biotransformed nor does it interact with carbon dioxide absorbents. However, desflurane is associated with troublesome cardiovascular stimulation involving tachycardia and both pulmonary and systemic hypertension. Sevoflurane appears to be advantageous for three reasons: firstly, because of its pleasant odour and consequent suitability for induction by inhalation, particularly in paediatric anaesthesia; secondly, it can be used with currently employed vaporizers, and thirdly, surgical demands can be met by lower doses, because its potency is higher.